

INTERPRETABILITY, STABILITY, AND ALGORITHMIC FAIRNESS

Term: Fall 2026	Instructor: Prof. Christophe Pérignon
Location: HEC Campus	Email: perignon@hec.fr

PROJECT DESCRIPTION:

Groups: Students will work in **groups of six**. Groups will be formed by the Student Representatives.

Goal: The goal of this project is to conduct and present a **scoring analysis** (binary target variable: 0/1) for a hypothetical client.

You will develop and compare three types of models:

1. One white-box model (e.g., logistic regression);
2. One machine-learning model (e.g., Random Forest, XGBoost);
3. One Tabular Foundation Model.

For each model, you will assess four dimensions:

- **Predictive performance**, considering both statistical and economic performance;
- **Interpretability**, including your ability to explain individual predictions and the main drivers of the model;
- **Stability**,
- **Fairness**, using appropriate fairness metrics and analyses.

Your analysis should go beyond simply reporting metrics. You should discuss the **trade-offs** between performance, interpretability, stability, and fairness across the three models.

Based on your findings, provide a **clear recommendation** to your client about which model should be deployed in production and why. Your recommendation should reflect the requirements of a trustworthy AI system rather than predictive performance alone.

Dataset: Each group is free to choose any dataset it finds interesting, provided that the prediction problem involves a **binary target variable** and is suitable for analyzing the four dimensions above.

Your proposed dataset must be submitted to the instructor for pre-validation by **Thursday, September 24, at 9:40 AM**.

Deliverables: Each group must deliver:

- A slide deck
- The code or a notebook containing the complete analysis, from data preparation to model evaluation;
- An application or dedicated website allowing the client to interact with, use, and test the different models.

All the deliverables have to be sent by email to the instructor by Monday September 28, at 9:40AM.

Presentation: Each group will present its project **on campus to a jury on Monday, September 28, between 9:40 AM and 5:50 PM**. The exact presentation time for each group will be confirmed later.

Each group will have:

- **15 minutes** for the presentation;
- **10 minutes** for Q&A.

Every team member must be fully prepared to answer questions about **any part of the analysis, code, models, and presentation**.

PROJECT EVALUATION:

The project evaluation criteria are:

Technical skills	(0 to 5 points)
Presentation skills	(0 to 5 points)
Q&A management	(0 to 5 points)
<u>Slides & Code/Appli/Notebook</u>	<u>(0 to 10 points)</u>
Total	(0 to 25 points).

Every team member must be fully prepared to answer questions on any section of the code and presentation. Note that grades can vary across team members.

Students receiving a non-passing grade will have the opportunity to take a 30-minute individual oral exam the following week, covering the entire course material as well as the project.